

The Utility Problem

Outages and reliability are one of the most important metrics for utility companies. Electricity outages have three large negative impacts on electricity distribution utilities: customers become dissatisfied with their service; regulators impose restrictions and fines for frequent and long-duration outages; and utilities lose revenue due to unused energy during the outage. Although there are different stakeholders for each of the three ramifications, these are all significant concerns for the utility as a whole.

Currently, depending on the utility, lost revenue is calculated either through estimates or is not calculated at all. In order to estimate the lost revenue, the utility first has to determine the customers who are affected by the outage. The utility then has to make estimates for the lost energy usage based on the date, time and historical data. In order to calculate SAIDI, SAIFI and other outage metrics that regulators are concerned with, they will either have to use a different outage management system or do another manual calculation.

These negative impacts put further strain on the corporation, requiring time to be focused on manual calculations rather than automating the solution. In addition, poor customer satisfaction can lead to reduced revenue and is becoming ever more threatening with distributed energy becoming more and more common. Finally, regulatory action will result in fines as well as negatively impact the reputation of the utility. Overall, outages are a very important issue for utilities and calculating unserved energy can help utility executives make decisions for future reliability upgrades.

The Data Solutions and Methods

An accurate estimate of the unserved energy due to an outage can be calculated using the Awesense Energy Transition Platform. Awesense Platform's unique linkage of geospatial data (GIS) with time-series data enables this type of analysis. With data populated in the Awesense Platform, the Energy Data Model's (EDM) database views and functions will accelerate the building of the analysis of unserved energy due to an outage. In order to achieve the desired analysis, the following data is required:

1. Outage Management System (OMS) data
 - a. Enables the utility to identify the location of the outage and/or set of affected consumers
 - b. Allows for predictions of future outages
2. Geospatial Information System (GIS) and Connectivity data
 - a. Along with Awesense's Energy Transition Platform, utilities can trace down the feeder to identify which meters are experiencing outages
3. Time-series (AMI/AMR/BILLING) data
 - a. Enables estimations for energy usage during outages.

All of this data is accessible through Awesense's EDM API. In EDM, the following methods will retrieve the required data from the datasets. The data ingested from the OMS is first queried and

Analysis of Unserved Energy due to Outage

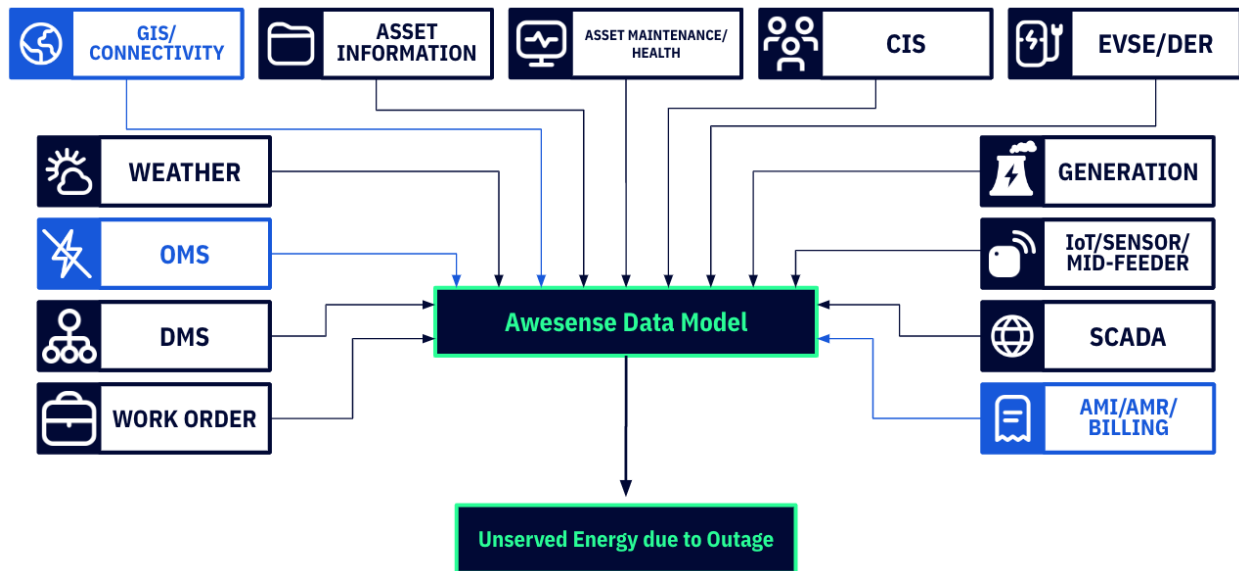
[UC11-01] - Awesense Energy Transition Platform Use Case



identified to locate the position of the outage on the grid or specify the group of outage-affected energy consumers. When all of the outage affected consumers are identified, EDM's predefined function, `ts_data_source_select()`, returns the time-series of all consumers for the desired time period (for example, an average of the consumption during the same hours the previous week). With the consumers and their corresponding estimated energy consumption, the entire unserved energy due to the outage can then be calculated. If desired, the utility can then multiply these by the time of use rates, which will provide the total monetary loss of the outage. Such a process can be automated, and historical or any future outage can be analyzed in terms of lost revenue due to unserved energy. Such insight should provide valuable information about where to spend resources.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.